

## Course Catalogue

| Module Code                        | Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ECTS | SWS | Lecture       | Tutorial     | Lab     |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|---------------|--------------|---------|
| MECH-B-5-ANT-ANT-ILV               | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 5    | 3   | 2             | 0            | 1       |
| <b>Course Name</b>                 | Drive Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |     |               |              |         |
| <b>Lecturer</b>                    | D. T. McGuiness, Ph.D (Daniel.McGuiness@mci.edu) (4A-434c)                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |     |               |              |         |
| <b>Study Programme</b>             | Mechatronic Design Innovation                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |     |               |              |         |
| <b>Official Name</b>               | Antriebstechnik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |     |               | <b>Lingo</b> | English |
| <b>Lecture Prerequisites</b>       | The student should have a good background in circuit analysis and familiarity with integral/differential calculus. After completion, the student will have a good grasp of electrical machines from their construction, how they are designed, what types out there and what applications are possible.                                                                                                                                                                                              |      |     |               |              |         |
| <b>Course Objectives</b>           | The goal of this lecture is to give an introduction to the fascinating world of electric machines. We will start from understanding how they work to the types currently used in commercial and industrial applications. The lecture will focus on mathematical and engineering concepts which will have direct use with real-life engineering applications.                                                                                                                                         |      |     |               |              |         |
| <b>Primary Course Content</b>      | Lecture Homepage on GitHub   WebBook                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |     |               |              |         |
| <b>Secondary Course Content(s)</b> | <i>Advanced electric drives: analysis control and modeling using MATLAB/Simulink</i> by Mohan Ned. , <i>Analysis of electric machinery and drive systems</i> by Krause Paul C. et. al. , <i>Design of rotating electrical machines</i> by Pyrhonen Juha et. al , <i>Electric Machinery Fundamentals (5th Edition)</i> by Stephen J. Chapman. , <i>Electric Machinery</i> by Fitzgerald A. E. et. al. , <i>Electric Motors and Drives: Fundamentals Types and Applications</i> by Hughes A. et. al. , |      |     |               |              |         |
| <b>Homework(s) and Project(s)</b>  | Personal Assignment (20) Final Exam (30) Lab Work (50)                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |     |               |              |         |
| <b>Assessment Criteria</b>         | <b>Assignment Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |     | <b>Effect</b> | <b>Count</b> |         |
|                                    | Personal Assignment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |     | 20            | 1            |         |
|                                    | Final Exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |     | 30            | 1            |         |
|                                    | Lab Work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |     | 50            | 4            |         |

## Lecture Structure

| Order | Topic                                                                                                                                                                                                                                                                                                                          | Units | Self Study |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------|
| 1     | Magnetic Circuits and Materials                                                                                                                                                                                                                                                                                                | 4     | 4          |
|       | <i>Introduction / Flux Linkage, Inductance, and Energy / Magnetic Material Properties / AC Excitation</i>                                                                                                                                                                                                                      |       |            |
| 2     | Transformers                                                                                                                                                                                                                                                                                                                   | 4     | 4          |
|       | <i>The Stationary Electric Machine / Transformers under No-Load Conditions / The Effect of Secondary Current on an Ideal Transformer / Reactances and Types of Equivalent Circuits / Understanding Transformer Behaviour / Auto and Multiwinding Transformers / Poly-Phase Transformers / Voltage and Current Transformers</i> |       |            |
| 3     | Electromechanical Energy Conversion                                                                                                                                                                                                                                                                                            | 4     | 8          |
|       | <i>The Necessary Definitions / Forces and Torque under Magnetic Field / The Theory of Energy Balance / Energy in a Singly Excited Magnetic Circuit / Determining Force and Torque from Energy / Determination of Magnetic Force and Torque from CoEnergy / Multiply Excited Magnetic Systems</i>                               |       |            |
| 4     | Lab 1: Installation of a Drive System                                                                                                                                                                                                                                                                                          | 3     | 6          |
|       | <i>Hands-on Laboratory Work</i>                                                                                                                                                                                                                                                                                                |       |            |
| 5     | Rotating Magnetic Fields                                                                                                                                                                                                                                                                                                       | 2     | 4          |
|       | <i>Fundamental Concepts / MMF of Distributed Windings / Magnetic Field in Rotating Machinery / Describing Rotating MMF Waves in AC Machines / Generated Voltage / Torque in Non-Salient-Pole Machines / Magnetic Saturation / Leakage Flux</i>                                                                                 |       |            |
| 6     | Lab 2: Building a Lathe                                                                                                                                                                                                                                                                                                        | 4     | 8          |
|       | <i>Hands-on Laboratory Work</i>                                                                                                                                                                                                                                                                                                |       |            |
| 7     | Synchronous Machines                                                                                                                                                                                                                                                                                                           | 4     | 8          |
|       | <i>To Spin with Synchronicity / Equivalent Circuits / Open and Short Circuit Characteristics / Steady-State Power-Angle Characteristics / Steady-State Operating Characteristics</i>                                                                                                                                           |       |            |
| 8     | Poly-phase Induction Drives                                                                                                                                                                                                                                                                                                    | 4     | 8          |
|       | <i>Introduction to Induction Machines / Current and Fluxes in Induction Machines / Induction Machine Equivalent Circuit / Analysis of the Equivalent Circuit / Calculating Power and Torque using Thevenin / Parameters Determination from No-Load and Blocked-Rotor Tests / The Effects of Rotor Resistance</i>               |       |            |
| 9     | Lab 3: Synchronous and Induction Drives                                                                                                                                                                                                                                                                                        | 4     | 8          |

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| Order | Topic                                                                                                                                                                                                                      | Units | Self Study |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------|
|       | <i>Hands-on Laboratory Work</i>                                                                                                                                                                                            |       |            |
| 10    | DC Drives                                                                                                                                                                                                                  | 4     | 8          |
|       | <i>Introduction   The Process of Commutation   Analytical Study   Armature Reaction   Steady-State Performance   Permanent Magnet Motors   Commutation and Interpoles   Compensation Windings   Series Universal Motor</i> |       |            |
| 11    | Special Machines                                                                                                                                                                                                           | 2     | 3          |
|       | <i>Single-Phase Induction Motors   Shaded Pole   Brushless DC Motors   Switched Reluctance Motors</i>                                                                                                                      |       |            |
| 12    | Lab 4: Photovoltaic Cells                                                                                                                                                                                                  | 4     | 8          |
|       | <i>Hands-on Laboratory Work</i>                                                                                                                                                                                            |       |            |
| 13    | Final Exam                                                                                                                                                                                                                 | 2     | 4          |
|       | <i>Written Exam</i>                                                                                                                                                                                                        |       |            |
| 14    | Total                                                                                                                                                                                                                      | 45    | 90         |
|       |                                                                                                                                                                                                                            |       |            |

- Any major announcements will be made on SAKAI regarding any possible date/content/structural changes for the assignment(s), exam(s).
- Any lecture material will be posted at the lectures corresponding GitHub home-page. The link will be present on the lectures SAKAI homepage.
- If there are any questions regarding course content/exams/assignments please do not refrain from contacting me (Daniel.McGuiness@mci.edu).